ABSTRACT OF THE DISCLOSURE

A method and system for encoding an analog signal directly into a sequence of high-cardinality data words for transmission or storage. The system bypasses the traditional step of analog-to-digital conversion into a binary format. Instead, samples of an analog waveform are directly mapped to discrete states in a high-radix, parallel data transmission system, such as wavelength-encoded states of a parallel optical interface. Each state, represented by an optical element with a specific wavelength, corresponds to a specific amplitude or characteristic of the analog signal at a sampling point. This architecture provides efficient representation of analog data, enabling high-density storage and parallel transmission.